Nokia Core Routing Protocol Flap SOP: ALARM_4938 on BLR-CORE-02

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Warning | **Alarm Code:** ALARM_4938

Description

This Standard Operating Procedure (SOP) addresses the failure context of Node ID: BLR-CORE-02, Site Location: Unknown Site, Vendor (CLI Dialect): Nokia, Role: Core, Failure Domain: Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding), Alarm Code: ALARM_4938, Severity: Warning. The underlying protocol or hardware mechanics causing this failure domain on this specific role and vendor equipment can be attributed to the inherent complexities of routing protocol convergence and the dynamic nature of BGP, OSPF, and ISIS routing protocols. In a Core node like BLR-CORE-02, BGP route churn can lead to an increase in routing table updates, resulting in potential BGP protocol instability. Similarly, OSPF adjacency loss and ISIS LSP flooding can lead to routing protocol instabilities, further exacerbating the scenario. Furthermore, hardware and software issues, including CPU congestion, memory issues, and software bugs, can also contribute to this failure domain. It is essential to identify the root cause and implement corrective actions promptly to prevent network instability and potential outages.

Pre-requisite Checks

- Verify that the Node ID: BLR-CORE-02 is accessible via SSH or console.

- Ensure that the Unknown Site is equipped with necessary infrastructure and support for remote hands, if required.

- Confirm that redundant link status is functioning correctly and there are no apparent hardware or software issues on the redundant links.

Potential Root Causes

Cause	Probability
BGP protocol instability due to excessive route churn	High
OSPF adjacency loss due to hardware issues or software bugs	Medium
ISIS LSP flooding due to hardware or software issues or configuration errors	Medium
CPU congestion or memory issues on the Node ID: BLR-CORE-02	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command is used to verify the BGP protocol stability and identify potential issues, such as excessive route churn.
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```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command is used to
verify OSPF adjacency status and identify potential issues, such as
adjacency loss.

Command: show isis database, Expected Output: Level-1 LSPs (3) and
Level-2 LSPs (1), last sequence 0x1, Description: This command is used
to verify ISIS LSP status and identify potential issues, such as LSP
flooding.
```

Resolution Steps

Step 1: Step: Verify BGP protocol stability, Command: clear bgp , Description: Clear BGP session on the Node ID: BLR-CORE-02 to resolve BGP protocol instability due to excessive route churn.

Step 2: Step: Verify OSPF adjacency status, Command: clear ospf neighbor , Description: Clear OSPF adjacency on the Node ID: BLR-CORE-02 to resolve OSPF adjacency loss due to hardware issues or software bugs.

Step 3: Step: Verify ISIS LSP status, Command: clear isis lsp , Description: Clear ISIS LSP on the Node ID: BLR-CORE-02 to resolve ISIS LSP flooding due to hardware or software issues or configuration errors.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 0,
main routing table version is 0, Description: Verify that the BGP
protocol stability has been restored after clearing the BGP session.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that the OSPF
adjacency status has been restored after clearing the OSPF adjacency.

Command: show isis database, Expected Output: Level-1 LSPs (3) and
Level-2 LSPs (1), last sequence 0x1, Description: Verify that the ISIS
LSP status has been restored after clearing the ISIS LSP.
```

Roll-back Procedures

- Step: Restore BGP session, Command: clear bgp , Description: Restore the BGP session on the Node ID: BLR-CORE-02 to the previous state before clearing it for troubleshooting purposes.

- Step: Restore OSPF adjacency, Command: clear ospf neighbor , Description: Restore the OSPF adjacency on the Node ID: BLR-CORE-02 to the previous state before clearing it for troubleshooting purposes.

- Step: Restore ISIS LSP, Command: clear isis lsp , Description: Restore the ISIS LSP on the Node ID: BLR-CORE-02 to the previous state before clearing it for troubleshooting purposes.